BerkeleyHaas

Master of Financial Engineering Program

A Reinforcement Learning Approach to Portfolio Optimization

MFE Spring 2022

Group 8

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March 09, 2022

Abstract

Portfolio optimization has been an active topic of research since the 1950s, to which particularly novel approach is Deep Reinforcement Learning (DRL). DRL has shown promising results in portfolio allocation [Alonso, Srivastava, et al. 2020]. We evaluate the efficacy and factor exposures of portfolios built with DRL asset allocations using Ken French's data [French n.d.] for 49 industry portfolios gathered from January 1, 1970 to December 31, 2021. Using the Advantage Actor Critic (A2C) algorithm we compare long-only DRL portfolios to three alternative long-only portfolio allocation strategies: mean variance, minimum variance and equal weighted. We find that DRL is a very computationally expensive approach. However, it offers diversification benefits when compared to mean variance and minimum variance portfolios. We also demonstrate that DRL is robust during crisis times.

Contents

1 Introduction and Hypothesis **8**

2 Alternative Portfolio Optimization Methods for Comparison **10**

2.1 Mean Variance Optimized Portfolio 10

2.2 Equal Weighted 10

2.3 Global Minimum Variance 10

3 Performance Measurement metrics **11**

4 Model Algorithms **11**

4.1 Reinforcement Learning Architecture: 11

4.2 Environment Implementation 13

4.3 Agent Implementation 14

4.3.1 Value Based Algorithms 15

4.3.2 Policy Based Algorithms 16

4.3.3 Model Based Algorithms 17

4.3.4 Hybrid - Advantage Actor Critic (A2C) Algorithms 17

5 Inputs **20**

6 Training the Model **20**

7 Backtesting **20**

8 Results **21**

8.1 Alternative Models 21

8.1.1 Mean Variance Portfolio 21

8.1.2 Minimum Variance Portfolio 22

8.1.3 Equal Weighted Portfolio 25

8.2 Deep Reinforcement Learning 26

8.2.1 Approach 1 26

8.2.2 Approach 2 26

8.2.3 Approach 3 27

8.2.4 Comparative study of the three DRL models 28

8.2.5 Factor exposures of DRL portfolio 31

8.3 Crisis Scenario Analysis 32

8.3.1 2008 Global Financial Crisis 32

8.3.2 2020 Covid Crash 36

9 Conclusion **40**

9.1 Summary 40

9.2 Future Work 41

List of Figures

1 RL model 12

2 Reinforcement Learning Agents Architecture 15

3 Actor Critic Architecture 18

4 Mean Variance weights 21

5 Minimum Variance weights 22

6 Returns Correlation for timeline 2001 to 2021 23

7 Returns Correlation for timeline 2004 to 2009 24

8 Equal Weights 25

9 Industry weights for Approach 1 26

10 Industry weights for Approach 2 27

11 Industry weights for Approach 3 28

12 Cumulative Log returns of 3 approaches vs S&P 500 29

13 Rolling Sharpe with a 6 month window of DRL approach 3 vs S&P 500 . . . 30

14 Log Cumulative Returns for Approach 3 30

15 Rolling betas of Carhart factors 32

16 Evolution of portfolio allocation across time during the Global Financial Crisis 33

17 Tilt of portfolio allocation during the Global Financial Crisis 34

18 Cumulative Log Returns of DRL compared with S&P500 and Minimum Variance during the GFC 35

19 Drawdowns of DRL compared with S&P 500 during the GFC 35

20 Evolution of Portfolio Weights across time during the Pandemic 37

21 Tilt of portfolio allocation in the crisis 38

22 Cumulative Log Returns of DRL compared with S&P 500 and Minimum Variance 39

23 Drawdowns of DRL compared with S&P 500 39

List of Tables

1 Summary Statistics for Mean Variance Portfolio 22

2 Summary Statistics for Minimum Variance Portfolio 23

3 Summary statistics for equally weighted portfolio 25

4 Summary Statistics 29

5 Summary Statistics 31

6 Carhart Model Factor exposures 31

7 Summary Statistics for DRL, Min Variance, and S&P 500 during the Global Financial Crisis 34

8 Fama French 4 factor loadings during the Global Financial Crisis 36

9 Summary Statistics for DRL, Min Variance, and S&P 500 during the Pandemic 38

10 Fama French 4 factor loadings during the Pandemic 40

1 Introduction and Hypothesis

Portfolio optimization, i.e. computing the optimal weights of assets in a portfolio for a given investment horizon, has been one of the fundamental challenges in the world of Finance. Over the past 70 years this area of research has evolved from simple static allocations like the 60/40 equity/bond portfolio to Modern Portfolio Theory (MPT) [Wikipedia 2022] and beyond. Traditionally, mean variance portfolio optimization is a multi-step process - estimating a vector of expected returns, then estimating a covariance matrix, and subsequently using these estimates as inputs to the portfolio optimizer. The minimum variance approach is similar but only uses the covariance matrix as inputs. While there have been some advances in estimating the covariance matrix accurately [Ledoit and Wolf 2003], estimation of the sample return vector still remains problematic. Therefore, the challenges are first an estimation problem for the covariance matrix and the expected returns (in the case of mean variance optimization), and then an optimization problem, typically a quadratic programming one.

In Markovitz's modern portfolio theory [Markowitz, 1952] the risk-adjusted return of a portfolio is maximized by mean-variance optimization (MVO). MVO theory aims to optimize a portfolio for a single time, thus based on a risk-return trade-off. The elegance of MPT belies a number of structural simplifications, namely considering only the first and second moments of the portfolio returns. Some academic researchers e.g. [Lee, 1977; Kraus and Litzenberger, 1976] have proposed alternate frameworks which include higher moments (skewness and kurtosis) or often a more realistic portrayal of the distribution of future portfolio returns. Moreover, MPT is sensitive to model inputs, and those model inputs are difficult to estimate. To counter this problem, practitioners have offered models that incorporate various simplifying assumptions. For example, the Black Litterman (BL) model [Wikipedia n.d.] estimates expected returns through assumptions of equilibrium and computes a weighted average of user-defined views and the equilibrium returns. In addition to the BL model, many other portfolio allocation methods like equally weighted, equal volatility, max Sharpe ratio, and risk parity have also been explored. One common feature among the methods listed is that information used in obtaining the weights is derived only from the in-sample data, i.e., past

returns.

In the era of super computers and big data, the use of machine learning and deep learning algorithms has shown outstanding success in areas such as image recognition and detection. However, given the complex nature of financial markets, applications to financial data impose a higher bar for these machine learning and deep learning algorithms, in part because when these algorithms are applied to live markets, the actions derived from these algorithms influence the market. Even so, in a supervised setting, researchers have demonstrated improved prediction accuracy by utilizing advanced deep neural networks such as Recurrent Neural Networks and Long Short Term Memory models [Noguer I Alonso et al. 2020]. Given these algorithms' ability to ingest large quantities of data, we propose to explore the idea of applying deep reinforcement learning for portfolio optimization using exogenous variables like technical indicators, various market level indicators, and other non-asset-level data.

We aim to address the following questions in our study:

- How do simple DRL (Deep Reinforcement Learning) algorithms for portfolio allocation compare to traditional portfolio optimization models?
- Does the performance of DRL models improve upon adding additional features like technical indicators?
- What are the exposures of simple DRL algorithms to the market, size, value and momentum factors?
- Does the DRL algorithm have the potential for superior performance during extremely turbulent times?

Section 2 describes alternative portfolio allocation strategies for comparison. Section 3 lays out the metrics which we track to gauge the performance of our algorithm and alternative allocation approaches. An explanation of the specific DRL algorithms is presented in section 4. Sections 6 and 6 describe our inputs and our approach to training the model. Section 7

defines our trading and backtesting approach. Finally in section 8, we report the results of our study.

2 Alternative Portfolio Optimization Methods for Comparison

2.1 Mean Variance Optimized Portfolio

MVO portfolios are highly sensitive to their inputs. The optimization will take the inputs as defining the entire universe of possibilities and allocate weights accordingly. This can result in very few assets getting most of the weight and many assets having zero weight. Research has shown that under certain conditions the MVO approach can produce more diverse portfolios [Green and Hollifield 1992]. We will compare the DRL model to the MVO portfolio without any conditions except the weights must be long only and must sum to one.

2.2 Equal Weighted

The most naive approach to portfolio allocation would be an equal-weighted portfolio. Here we divide the fund equally among the available assets such that the weight of each asset is 1/n for n assets in the portfolio. This approach does not require any estimation of returns, risk or covariance matrices. Equal-weighted portfolios have attracted attention historically as they have often outperformed more advanced techniques [DeMiguel et al. 2007]. This outperformance could potentially be attributed to higher allocations to small-cap stocks which have exhibited higher returns (and volatility) in the past.

2.3 Global Minimum Variance

Mean variance optimization (MVO), which is derived from Modern Portfolio Theory (MPT), aims to optimize the portfolio for a single period by maximizing the expected return for a

given level of risk. It takes the expected return vector and covariance matrix as its inputs. However, Modern Portfolio Theory, and consequently MVO, has several flaws. MPT assumes that all investors are risk-averse and given a chance they will always select the portfolio that optimally balances risk vs reward, which is not always true in practice. The biggest challenge is the accuracy of data used in the optimization. The returns, risk, and covariance matrix calculated from historical data are not always stable. However, even with all its shortcomings, the Global Minimum Variance portfolio is widely used in practice. For this reason we chose it as an alternative portfolio:

$$w_{GMV} = \frac{1}{1'\Sigma^{-1}\mathbf{1}}\Sigma^{-1}\mathbf{1} \tag{1}$$

where Σ is the covariance matrix and $\mathbf{1}$ is vector of ones of size n.

3 Performance Measurement metrics

One of the critiques of the MPT portfolio theory is that it only takes into account the mean and covariance matrix and does not at all use drawdowns. [Investopedia n.d.] We will test a broad variety of metrics including annual returns, monthly returns, cumulative returns, annual volatility, Sharpe ratio and max drawdown to see if DRL outperforms traditional methods.

4 Model Algorithms

4.1 Reinforcement Learning Architecture:

Deep Reinforcement Learning (DRL) is a sub-field of machine learning, but has some key differences compared to traditional supervised or unsupervised learning which are discussed below. In DRL, we train an artificial intelligence called the 'Agent' to perform tasks by

interacting with a dynamic 'Environment'. A simple architecture of a Reinforcement Learning model is shown in Figure 1.

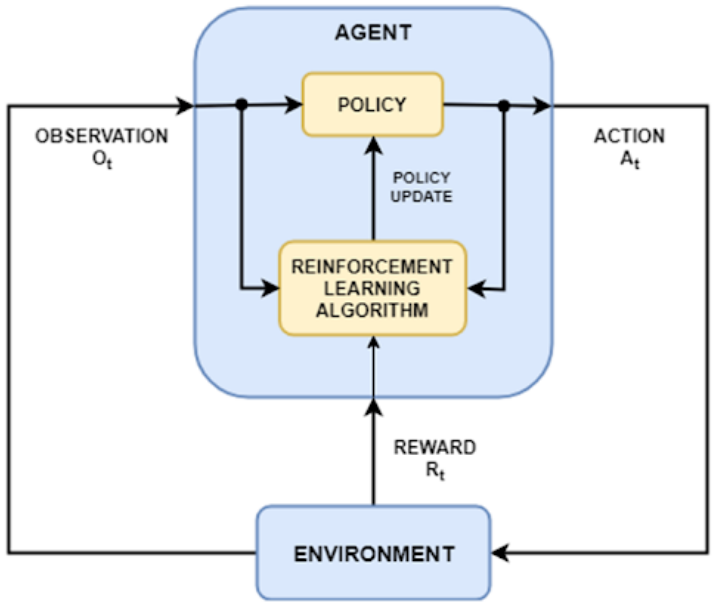

Figure 1: RL model
Source:(mathworks.com)

Below are the main components of a Reinforcement Learning model:

- **Agent**: This is the intelligence that we're going to train using observations from the environment. It chooses from a set of available actions to increase possible rewards.
- **Environment**: This is the unknown dynamic world with which our agent interacts.
- **Observation**: Current state of the Environment as seen by the Agent
- **Actions**: The set of different moves an Agent can take
- **Policy:** The Agent uses the policy to choose Actions in response to the State/Observation. In RL, we try to optimize this policy over time

A training loop consists of the following steps:

1. The Environment provides the Agent with an Observation (or State), O_t. This is what the Agent observes about the Environment based on its current State.

2. Based on the State/Observation, the Agent determines what Action it should take. This should put the Agent in a new State.

3. Additionally, the Action affects the current Environment, and the feedback from the Environment is passed back to the Agent in terms of a Reward. The Reward can be positive or negative.These Rewards are determined at each time step.

4. Based on the Rewards being positive or negative, the Agent's Actions are reinforced. The Agent learns whether the Action taken was good or bad, and hence updates its Policy accordingly.

5. This process repeats dynamically, and Agents typically try to maximize expected Rewards over time.

For our portfolio optimization problem, the Environment will be a simulation or state of the stock market and the Agent would act as a trader. The state would consist of the current statistics about the market - the previous day's open, high, low, and close (OHLC), technical indicators and the covariance matrix between the asset returns. Our actions would be the choice of portfolio weight allocation and our reward would be the portfolio value. The policy would be the probability distribution of portfolio weight actions at a particular state determined by the covariance matrix and other market variables.

4.2 Environment Implementation

In our implementation we plan to create a custom portfolio allocation environment using Open AI's gym package [AI n.d.] In designing the Environment, we simulate how a human trader would perceive and process the information to make portfolio allocation decisions. The Environment would possess the necessary features to enable the Agent to learn a suitable policy aimed at maximizing rewards.

- **Action Space:** would contain the set of all the possible actions the agent can take. In

our case, it would be the portfolio weights of the assets ranging from 0 to 1 with the constraint that they sum up to 1.

- **Observation space/State:** would be defined to contain all the environment's data available to the Agent before deciding on an action. In our implementation , the state space would constitute the previous day's OHLC prices, covariance matrix of the assets looking back up to 1,250 trading days (five years), and technical indicators. The underlying idea is that as a human trader considers different information before executing a trade, the Agent should also be able to learn to use this information to identify the optimal portfolio weights.

- **Reward:** The reward function is the portfolio value after taking the action and observing the next period market returns.

4.3 Agent Implementation

There are 3 main types of Agent Implementations in Deep Reinforcement Learning:

1. Value Optimization
2. Policy Optimization
3. Model Optimization

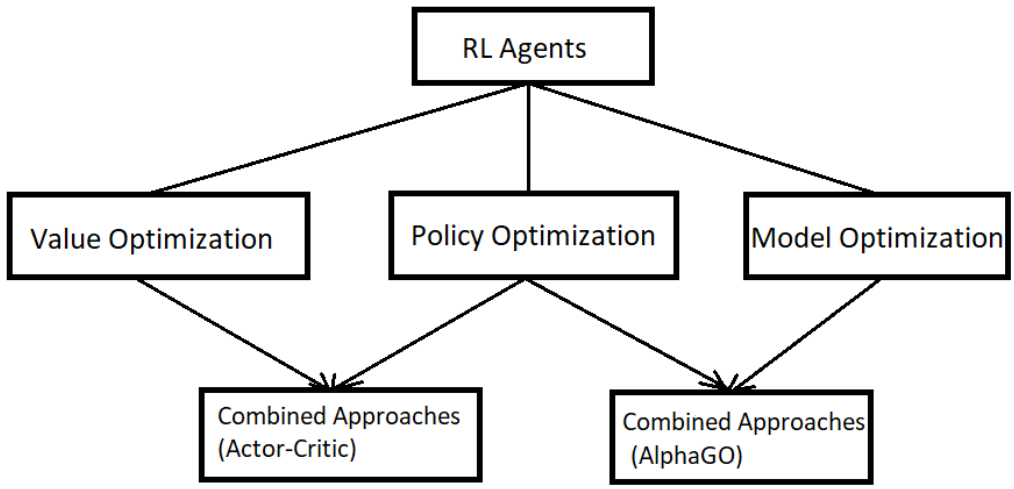

Figure 2: Reinforcement Learning Agents Architecture

Different Reinforcement Learning implementations use anyone or a combination of these approaches (See Figure 2 for a diagram showing different Reinforcement Learning agent architectures).

4.3.1 Value Based Algorithms

In the Value based algorithms, we solve a reinforcement learning problem by optimizing the Value Function, $V_\pi(S)$. A Value Function is the expected value of future rewards. At each state, the value would be the total discounted rewards the agent expects to get, starting at that state. The agent hence uses this function to decide the next action such that it gets the maximum expected value:

$$V_\pi(s) = E_\pi[\underbrace{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + ...}_{\text{Discounted Future Rewards}} | \underbrace{S_t = s}_{\text{Given the State S}}]$$

where,

R_{t+1} = Reward at time t+1,

γ = discount factor

These algorithms are simple to implement and make efficient use of training data. But if the state-action pair becomes large, the value function can become highly complicated and

sometimes intractable to estimate the optimal value. This is because their predictors try to assign a value to each of the possible actions at the current state, which becomes difficult when there is a continuous action space.

4.3.2 Policy Based Algorithms

The primary goal of the agent in any Reinforcement Learning algorithm is to learn a policy, π which maps the states to actions. In Policy based algorithms, the agent directly learns the policy function, π. This contrasts with the Value optimization algorithms, wherein the policy is indirectly learned by first estimating a Value function. The policy could be either Deterministic or Stochastic.

- **Deterministic:** It is used in deterministic environments. The policy function would return a predetermined action (a) for each given state (s).

$$a = \mu(s)$$

- **Stochastic:** The probability function would return a probability distribution of actions at any given state. This is used when the environment is uncertain. For our portfolio allocation task we would use a Stochastic policy since our environment is dynamic and has a continuous action-state space.

$$\pi(a|s) = \mathbb{P}[A_t = a|S_t = s]$$

Policy-based algorithms tend to follow the gradients to search for optimal policy and hence show smoother convergence characteristics. They are also more effective when there are large numbers of state-action pairs or a continuous action space. One major drawback is that they can take longer to converge or can converge to a local maximum [Simonini n.d.(b)].

4.3.3 Model Based Algorithms

In model-based Algorithms, the agent indirectly learns the optimal behavior by learning a model of the environment. The learned model predicts the outcomes of actions rather than interaction with the environment to learn optimal policies [Ray and Tadepalli 2010].

4.3.4 Hybrid - Advantage Actor Critic (A2C) Algorithms

The Actor Critic Algorithm is a hybrid of Value based and Policy based algorithms. It consists of an 'Actor' and a 'Critic' implementation with features as below:

- **Actor:** An Actor that controls how the Agent behaves and decides on which Actions to take. It's implemented based on a Policy based algorithm.
- **Critic:** As the name suggests, a Critic evaluates the Actor's actions and provides feedback on how to make adjustments. It is implemented using Value based algorithms.

The advantage of using a hybrid approach is that it overcomes the disadvantages of each of the individual algorithms [Karagiannakos n.d.] Hence, it can be applied more efficiently to a wider range of problems than a Value based algorithm and has lesser variance than a Policy based approach.

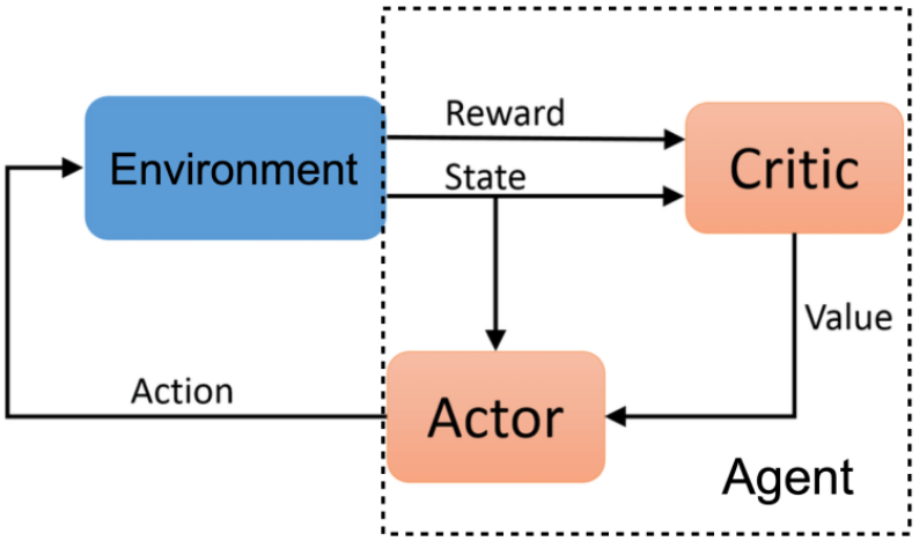

Figure 3: Actor Critic Architecture
Source:(ResearchGate)

Figure 3 shows the architecture of an Actor Critic Agent. The Actor and Critic interact with each other during the training period in the following ways:

- At each time step t, the Actor takes the current state (S_t) from the Environment as input.
- The Actor, based on the Policy learned so far, outputs the best Action (A_t).
- The Critic receives the Reward from the Environment and the Action by the Actor and computes the value of taking that Action at that State.
- The Actor receives this feedback from the Critic and updates its Policy function parameters.
- The Actor and Critic are implemented as two neural networks. They are trained separately but synchronously. They learn at the same time, constantly interacting. The

actor is continually changing the policy to exceed the critic's expectation, and the critic is constantly updating its value function to evaluate the actor's changing policy.

- As time passes, the Actor learns to produce better and better actions (as it learns the policy in a better manner) and the Critic gets better at evaluating those actions.

In the Advantage Actor Critic (A2C) approach, an advantage function, $A(s, a)$ is calculated by the Critic at each step instead of the Value function. The Advantage function is defined as [Simonini n.d.(a)]:

$$A(s, a) = Q(s, a) - V(s)$$

In a discrete time setting, the Q value, $Q(s, a)$ is a table of values for every state (row) and action (column) possible in the environment. Within each cell of the table, we learn a value for how good it is to take a given action within a given state. In a continuous time-frame, neural networks are used to implement a Q-Value function approximator. The V-value is the average value of a state.

Thus, intuitively, the advantage function tells us how much better it is to take a specific action than the average general action at the given state. In other words, this function calculates the extra reward the Actor gets if it takes this action. The extra reward is beyond the expected value of that state. If the action result is better than the expected, the actor is encouraged to update its policy parameters such that it increases the probability of that action. If an action result is less than expected, the actor is encouraged to adapt its policy such that probability of that action is reduced [Wang n.d.]

The A2C algorithm is most useful if we are interested in continuous action spaces because we reduce the policy variance by using the advantage function and update the policy in real time [Zhang et al. 2020]. Hence, for our portfolio allocation problem, we have implemented our agent using the A2C architecture. In the next section, we describe the key inputs which goes into our model implementation.

5 Inputs

We use industry (sector) portfolio data from the Ken French website [French n.d.] as our base input. We have daily returns data from January 1970 until December 31, 2021 for 49 equally weighted industry portfolios (See Appendix for details on the constituents of the Industry Portfolio). At each time step (daily) we calculate the covariance matrix of the returns using a 5 year look back period. We also augment our study with the use of technical indicators by constructing a price series from the return data.

6 Training the Model

We take three different approaches to train the model. These approaches are designed to test the hypotheses that 1) Retraining the model every year compared to training the model just once gives better results 2) Adding information like technical indicators to the model enhances the returns.

- Approach_1: Train from January 1970 to December 2000
- Approach_2: Expanding the training window for each year of trading. For example, trading for the year 2014, will involve training the model from 1970 to 2013.
- Approach_3: Approach_2 with technical indicators.

7 Backtesting

We will mainly use Quantopian's library of Pyfolio [Quantopian n.d.] to backtest our performance as it has a wide range of performance metrics and plots. As part of the backtest, we will start with an initial value of $1,000,000 and an equally-weighted portfolio. We calculate the stock returns and the updated portfolio value at each time-step. To realistically simulate the market conditions we need to include transaction costs incurred while trading. We

will add transaction cost as a per share percentage of 10 bps (1/100 of a percentage point). The transaction costs used would normally be a function of the size the portfolio and how often we are rebalancing. We chose 10 bps for simplicity. In the next section we discuss the backtesting results and compare it with alternative portfolio optimization models.

8 Results

8.1 Alternative Models

The alternative models we have chosen are mean-variance, minimum variance, and equally weighted. The results over the trade period of January 3, 2001 to December 31, 2021 are discussed in the following sections.

8.1.1 Mean Variance Portfolio

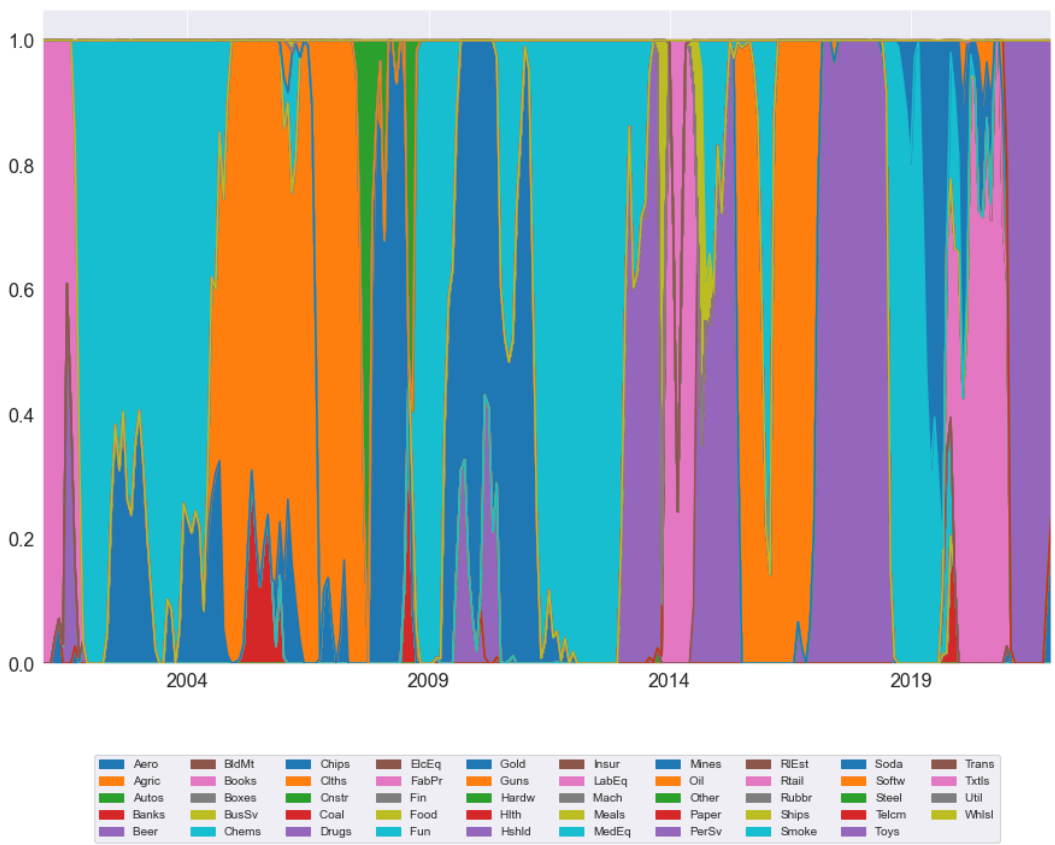

Figure 4: Mean Variance weights

The mean variance portfolio is highly sensitive to the input covariance matrix and forecast of returns. We chose to use the same window as the covariance matrix (past five years of data) for the expected returns using the historical mean. Industries having high mean and lower variance during that timeline gain the majority of the weight, leading to less diverse portfolios. Figure 4, shows the weights over time and we can see that mean variance optimiser takes extremely concentrated bets. Table 1 shows the summary statistics of our mean variance portfolio.

Strategy	Mean Return	Standard Dev	Max Drawdown	Sharpe Ratio	Daily Turnover	Alpha	Alpha P-value
Mean Variance	0.1273	0.2415	-0.6758	0.4677	0.0311	0.0002	0.1293

Table 1: Summary Statistics for Mean Variance Portfolio

8.1.2 Minimum Variance Portfolio

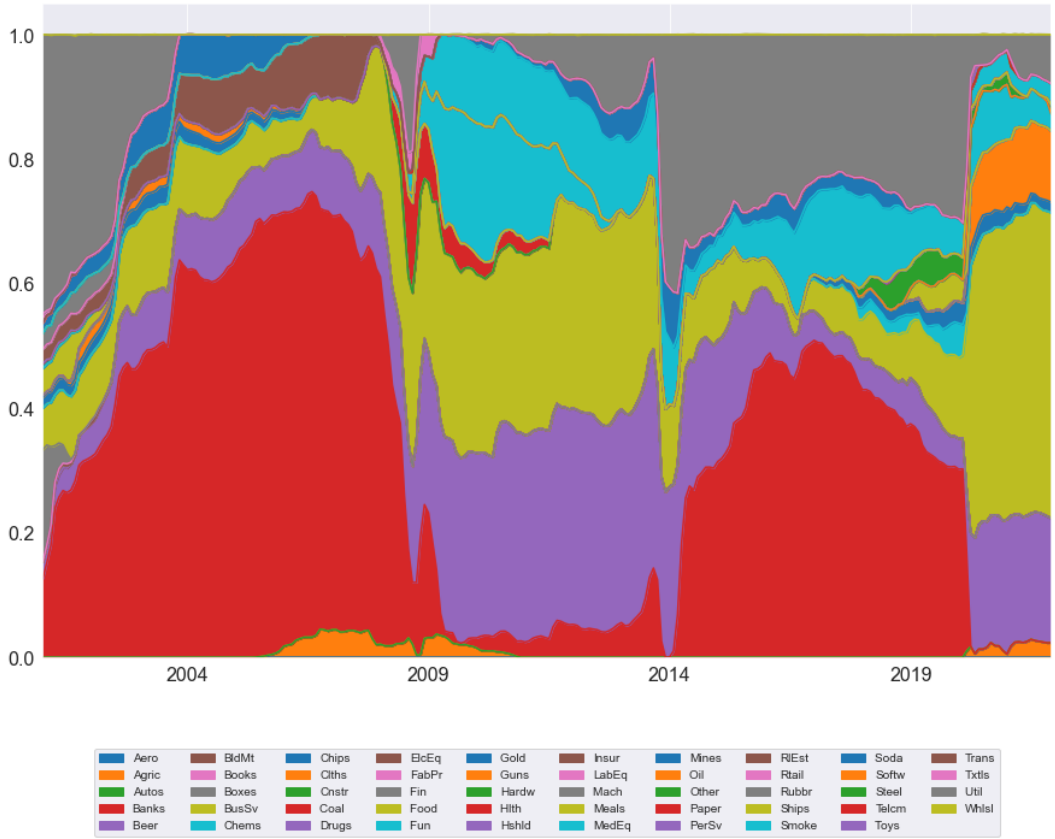

Figure 5: Minimum Variance weights

The minimum variance portfolio is also very sensitive to the inputs, although the minimum variance portfolio only uses the covariance matrix and does not use any expectation of returns. We again used the past five years equal weighted covariance matrix. The model allocates highly concentrated weights to sectors having low variance, namely Banks , Food , Utility and Beer. This strategy has the highest Sharpe ratio of the three alternative allocation strategies at 1.07. Table 2 shows the summary statistics of the Minimum Variance Portfolio.

Strategy	Mean Return	Standard Dev	Max Drawdown	Sharpe Ratio	Daily Turnover	Alpha	Alpha P-value
Minimum Variance	0.1721	0.1468	-0.5264	1.0702	0.0046	0.0004	0.0

Table 2: Summary Statistics for Minimum Variance Portfolio

Both the mean-variance model and the minimum variance model have very low diversification. To understand this better, we conducted a correlation analysis among the industries and present our findings in Figure 6 and Figure 7.

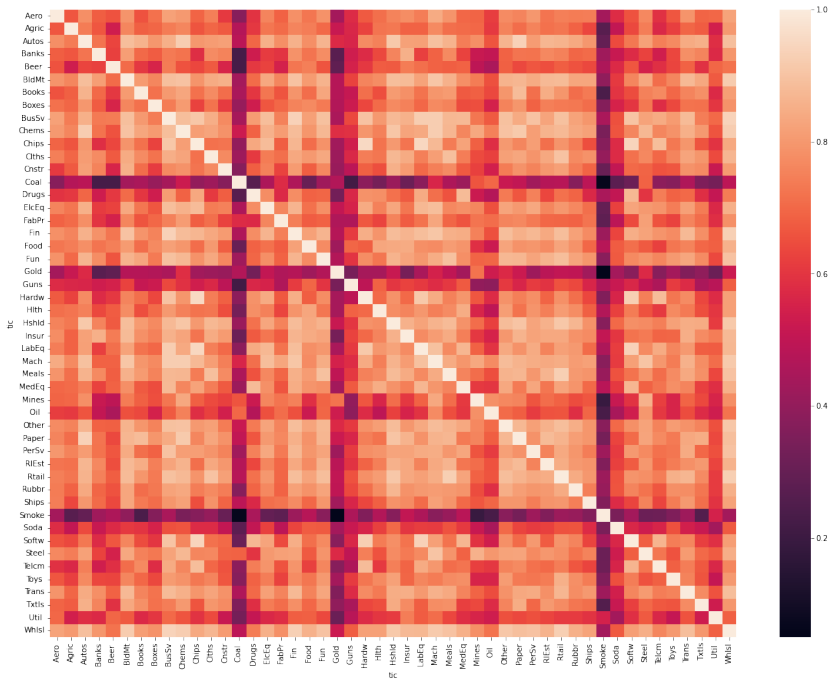

Figure 6: Returns Correlation for timeline 2001 to 2021

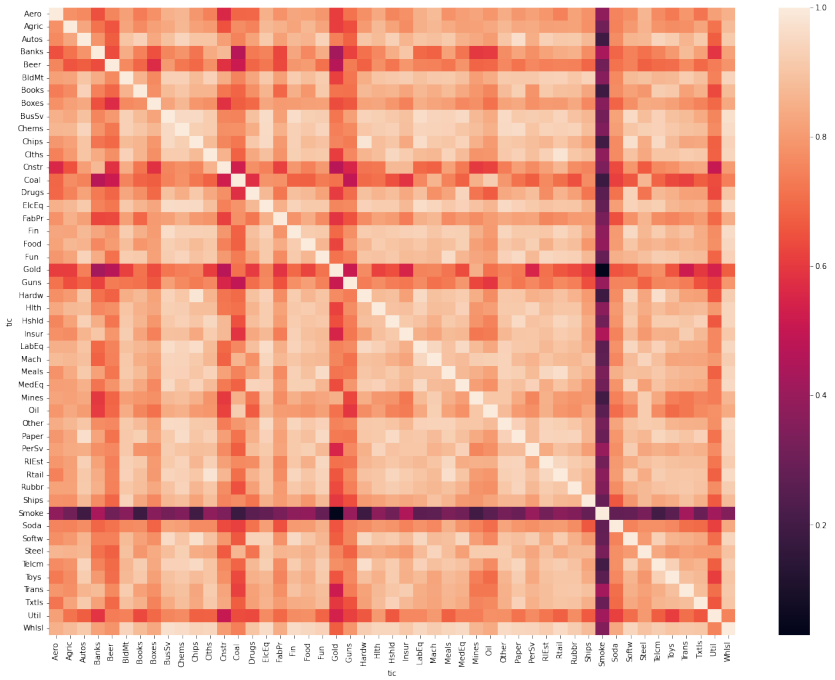

Figure 7: Returns Correlation for timeline 2004 to 2009

As we can see from the above heatmaps, there is high correlation among industries and thus, diversification has muted benefits.

8.1.3 Equal Weighted Portfolio

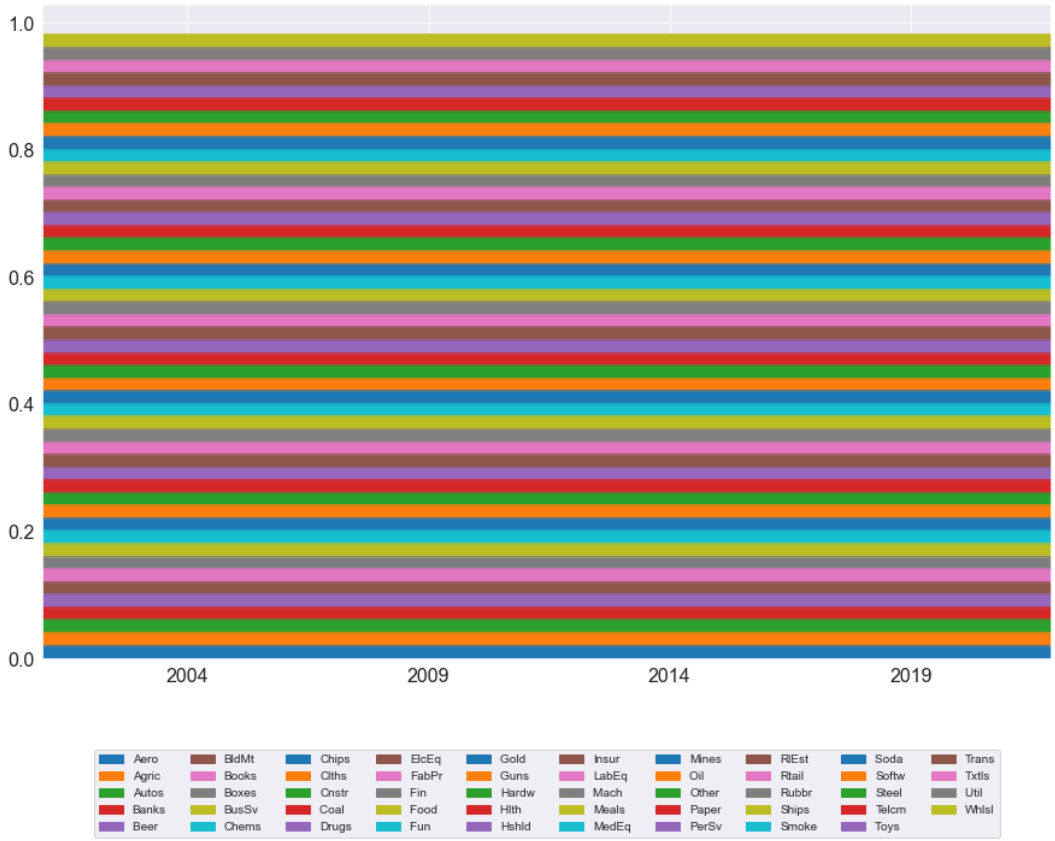

Figure 8: Equal Weights

The equal weighted portfolio naturally is much more diversified than both the minimum variance and mean variance portfolios. Table 3 shows summary statistics for Equally Weighted portfolio.

Strategy	Mean Return	Standard Dev	Max Drawdown	Sharpe Ratio	Daily Turnover	Alpha	Alpha P-value
Equal Weighted	0.1757	0.2032	-0.5946	0.7906	0.0033	0.0003	0.0

Table 3: Summary statistics for equally weighted portfolio

8.2 Deep Reinforcement Learning

8.2.1 Approach 1

In our first approach, as discussed earlier, we train our model from January 1970 to December 2000. We discuss in detail the performance of this approach in section 8.2.4, but it is interesting to see in the figure 9 how the weights evolve over time for this approach. Compared to Approach 2 and Approach 3, the weights generally do not change much over time except for 2008, 2014 and 2020 when there are shocks in the model due to the Great Financial Crisis (GFC), Ruble crisis and Covid.

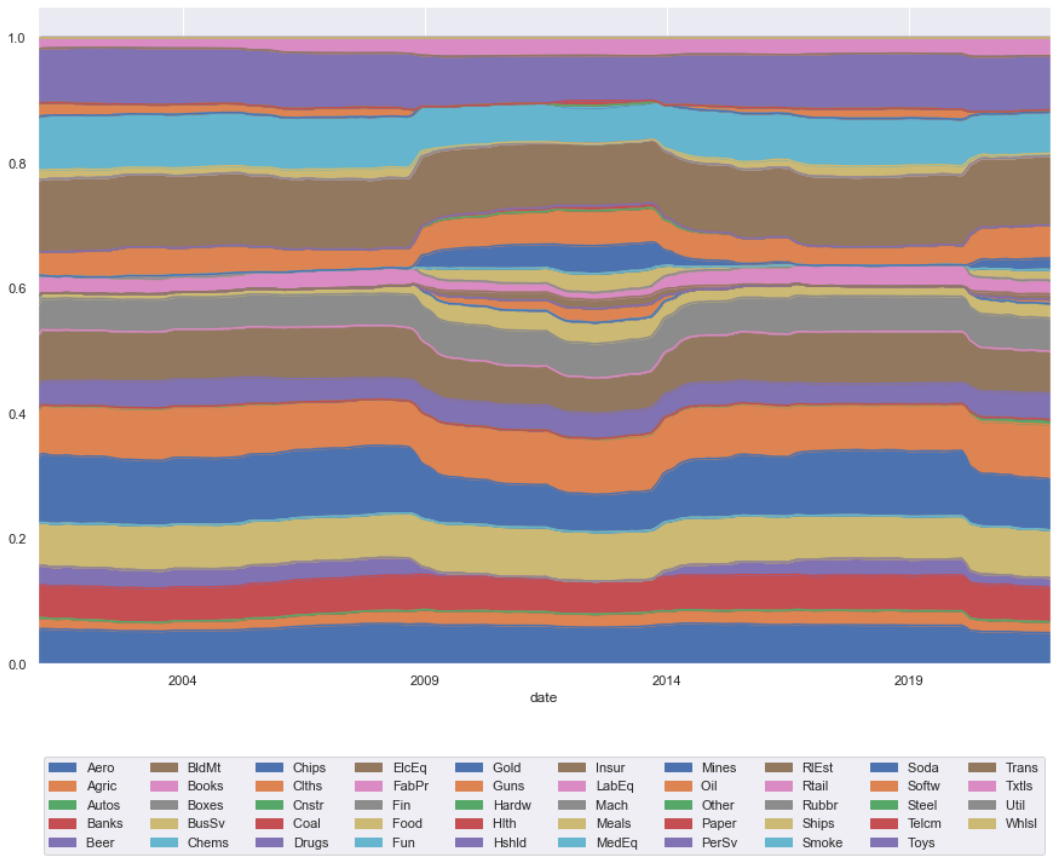

Figure 9: Industry weights for Approach 1

8.2.2 Approach 2

In this Approach, we train our model every year before we use it to trade for that year. In contrast to Approach 1, where we use one trained model to trade for the rest of the periods,

here we are using a new model for each year, while increasing the training window by one year. Our hypothesis is that the Approach 2 should perform better than Approach 1, since here we are training the model with additional most recent data. The inputs to this model is the returns time series and covariance matrix with 5 years lookback. We haven't added any user defined variables or Technical indicators in this approach as we wanted to learn the incremental benefits of providing additional data to the reinforcement learning model.

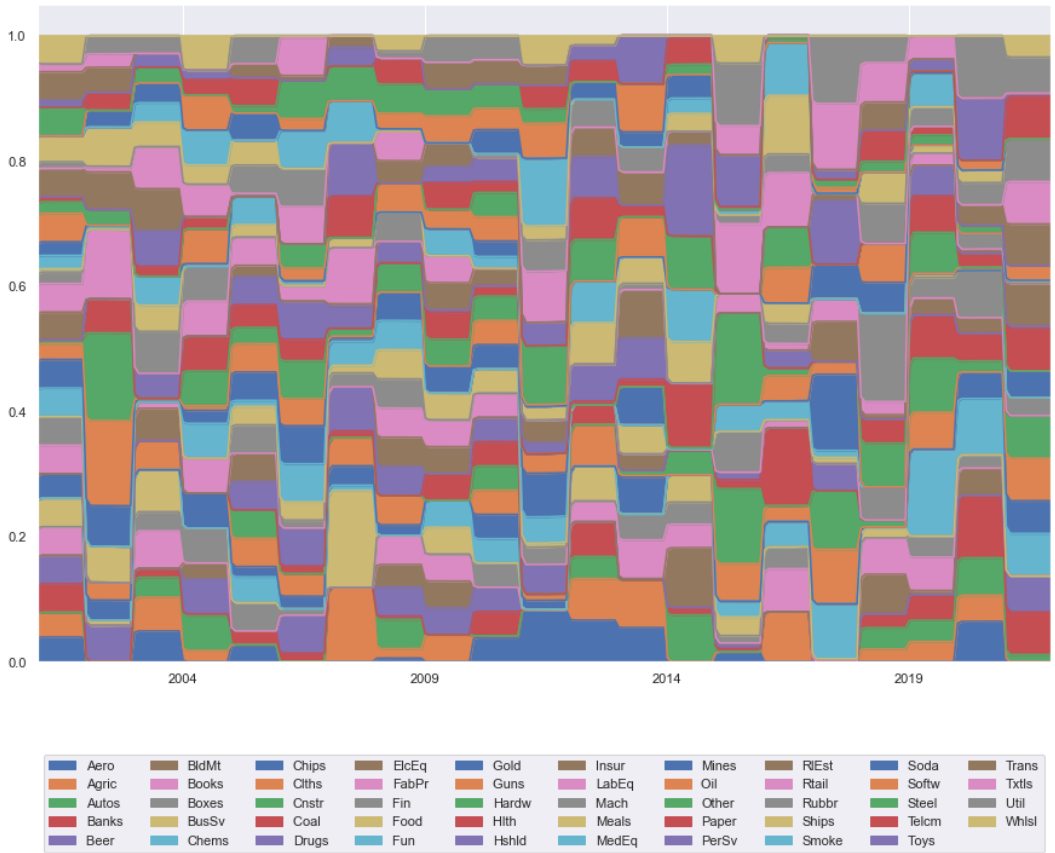

Figure 10: Industry weights for Approach 2

8.2.3 Approach 3

In this experiment, we train our model every year before we use it to trade for that year like an expanding window. But in contrast to approach 2, we provide our model with few technical indicators information. The technical indicators that we consider here are: "macd", "boll_ub", "boll_lb", "rsi_30", "cci_30", "dx_30", "close_30_sma", "close_60_sma". The intuition behind this approach is to show how a DRL model can be fed with additional

useful information such that it becomes more "market-aware". In a separate research [Koratamaddi et al. 2021] shows how one can incorporate market sentiment into a deep reinforcement learning model to improve their returns.

Initially, our reinforcement learning model was set to start trading with an equally weighted portfolio and adjust its weights depending on the change in market conditions. However, in approach 2 and approach 3 we retrain our model every year and trade for one year. So for these approaches we set our initial weights of our trading period same as the weights of last trading day of the previous period to maintain continuity.

We discuss the results of these three approaches in detail in the next section.

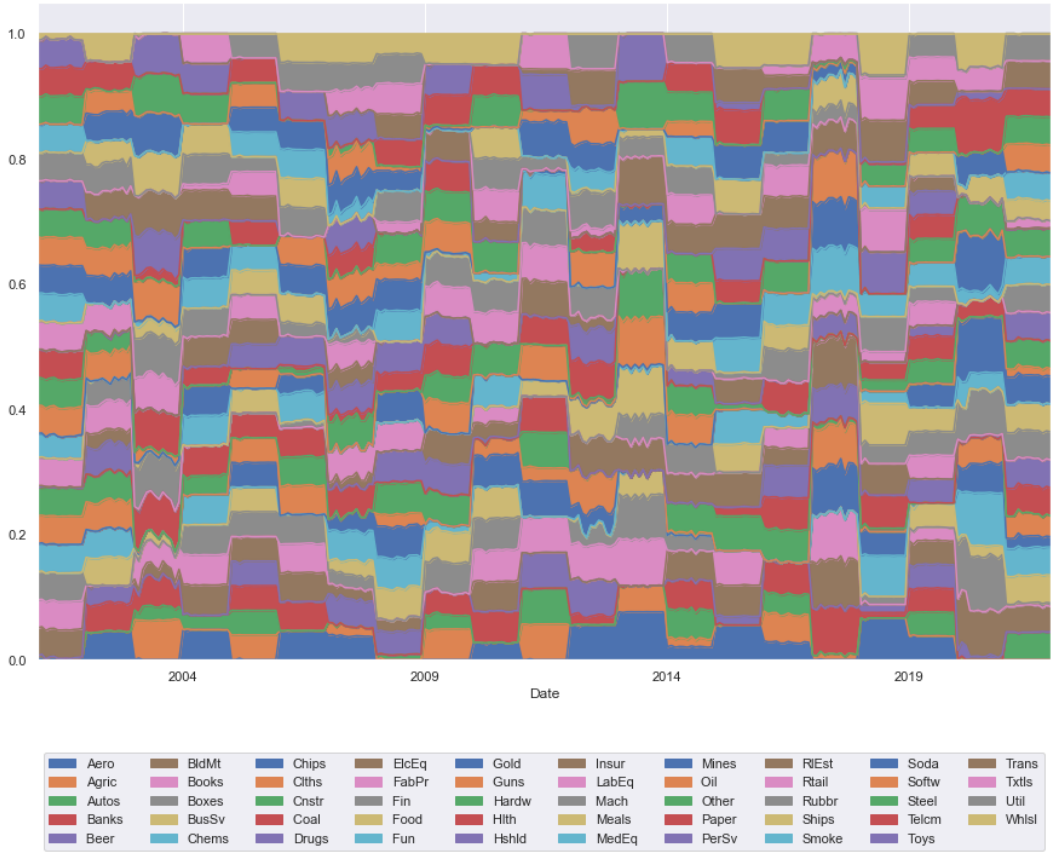

Figure 11: Industry weights for Approach 3

8.2.4 Comparative study of the three DRL models

We use the metrics mentioned in Section 3 to compare the three models introduced above and present summary statistics for each approach in Table 4:

	Approach_1	Approach_2	Approach_3	S&P 500
Annual return	0.159	0.165	0.173	0.065
Cumulative returns	20.903	23.724	27.198	2.724
Annual volatility	0.216	0.205	0.203	0.195
Sharpe ratio	0.789	0.849	0.886	0.419
Max drawdown	-0.607	-0.614	-0.585	-0.568
Sortino ratio	1.117	1.199	1.248	0.588
Daily value at risk	-0.027	-0.025	-0.025	-0.024

Table 4: Summary Statistics

Figure 12 and Figure 13 show the cumulative log returns and rolling sharpe ratios of the three approaches respectively.

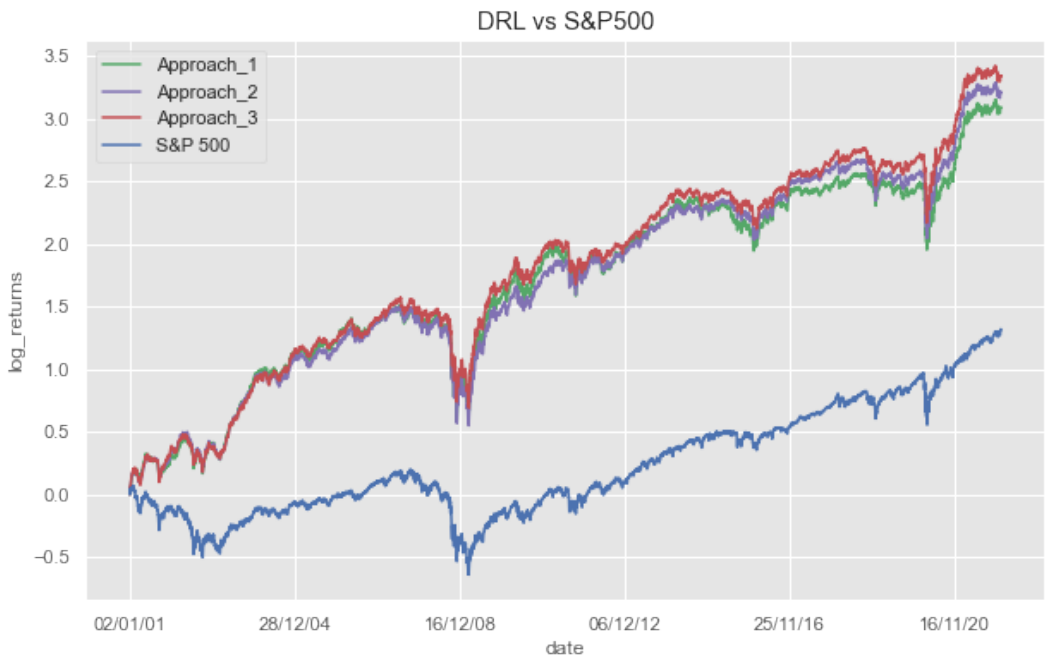

Figure 12: Cumulative Log returns of 3 approaches vs S&P 500

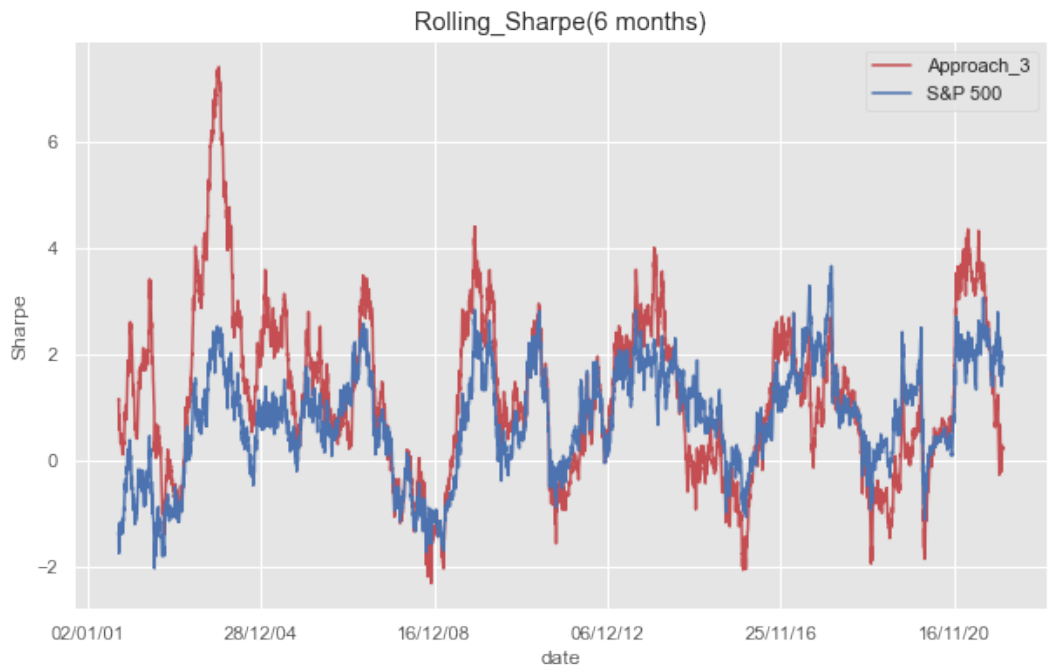

Figure 13: Rolling Sharpe with a 6 month window of DRL approach 3 vs S&P 500

Final cumulative return results for the third approach with DRL in red is shown below in Figure 14 :

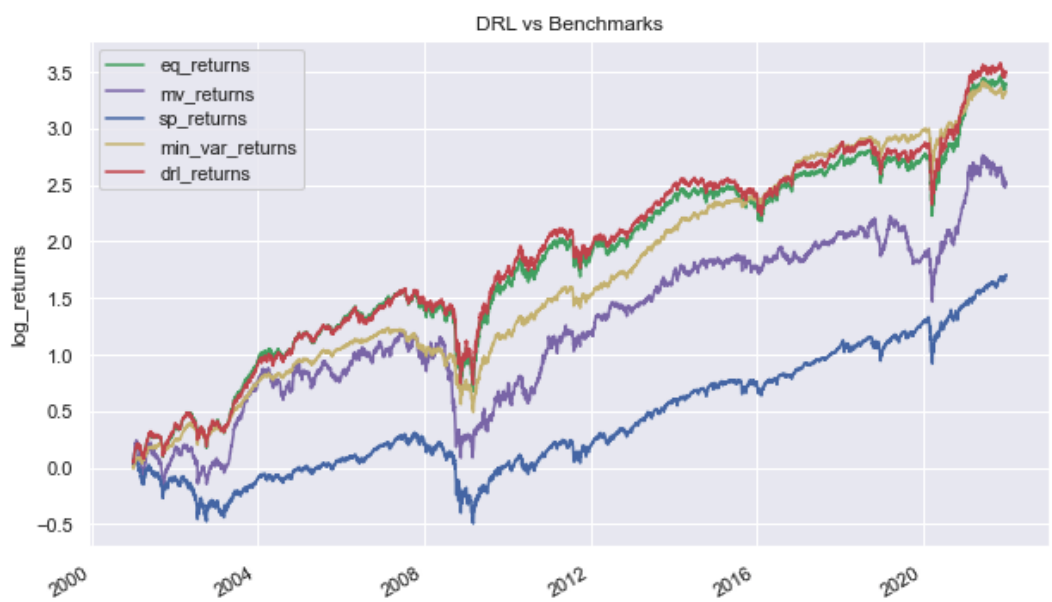

Figure 14: Log Cumulative Returns for Approach 3

Clearly, Approach 3 outperforms, the previous approaches. This is inline with our initial

hypothesis that retraining the model often and adding relevant market data to DRL enhances the performance of the model.

Table 5 shows how performance of Approach 3 stacks up against the traditional approaches.

Strategy	Mean Return	Standard Dev	Max Drawdown	Sharpe Ratio	Daily Turnover	Alpha	Alpha P-value
Equally Weighted	0.1757	0.2032	-0.5946	0.7906	0.033	0.0003	0.0000
Mean Variance	0.1279	0.2415	-0.6758	0.4699	0.0311	0.0002	0.1272
DRL	0.1817	0.2037	-0.5717	0.8175	0.038	0.0003	0.0000
S&P500	0.0845	0.1941	-0.5519	0.3641	NaN	-0.0000	0.4982
Min Variance	0.1722	0.1468	-0.5264	1.0710	0.0046	0.0004	0.0000

Table 5: Summary Statistics

Performance of our DRL model has a Sharpe Ratio is similar to Minimum Variance(MV) approach while also being more diversified than MV.

8.2.5 Factor exposures of DRL portfolio

We use the Carhart four-factor model [Carhart 1997] to measure the exposures of the DRL Approach 3 portfolio to HML, SMB, Mkt and Mom.

	Betas	T_stats
const	0.000282	5.553546
Mkt-RF	0.855998	196.805960
SMB	0.656953	77.666408
HML	0.280843	37.066839
Mom	-0.020323	-3.653471

Table 6: Carhart Model Factor exposures

The results shown in the above table are consistent with our intuition. The DRL portfolio has highly positive betas to "Mkt-RF", "SMB" and "HML".

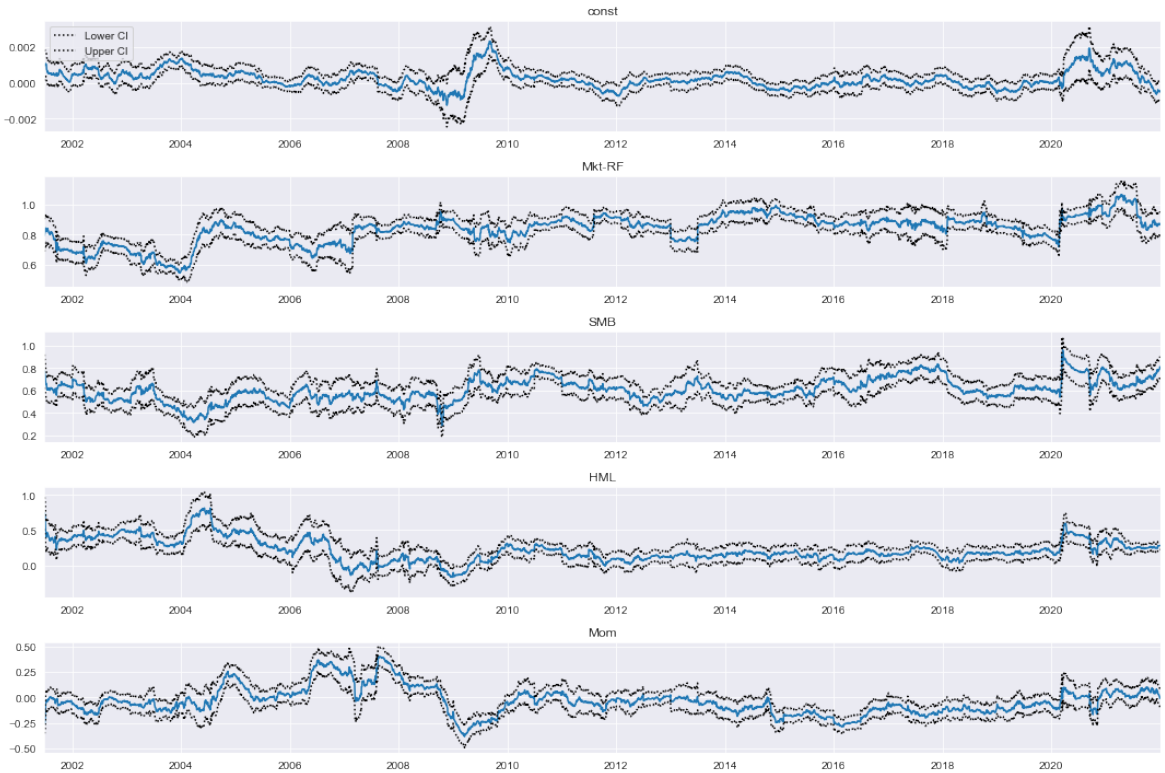

Figure 15: Rolling betas of Carhart factors

Figure 15 shows the 6 month rolling betas over time. Most notably, the exposures to Mkt-Rf and SMB never go below zero and alpha of this strategy is mostly zero except for times of crisis when our strategy seems to be out-performing and giving abnormal returns during this period. We delve deeper into it in the next section.

8.3 Crisis Scenario Analysis

8.3.1 2008 Global Financial Crisis

For this study we train the model from 01-01-1970 to 12-31-2006. We use this trained model to trade from 01-01-2007 to 12-31-2009. We choose this date range for training because we want the model to look at the full range of the data to best prepare for the crisis. Our prior hypothesis before using our model is that it should tilt away from the under performing industries. After running our model, we arrive at the time series of weights shown in Figure 16.

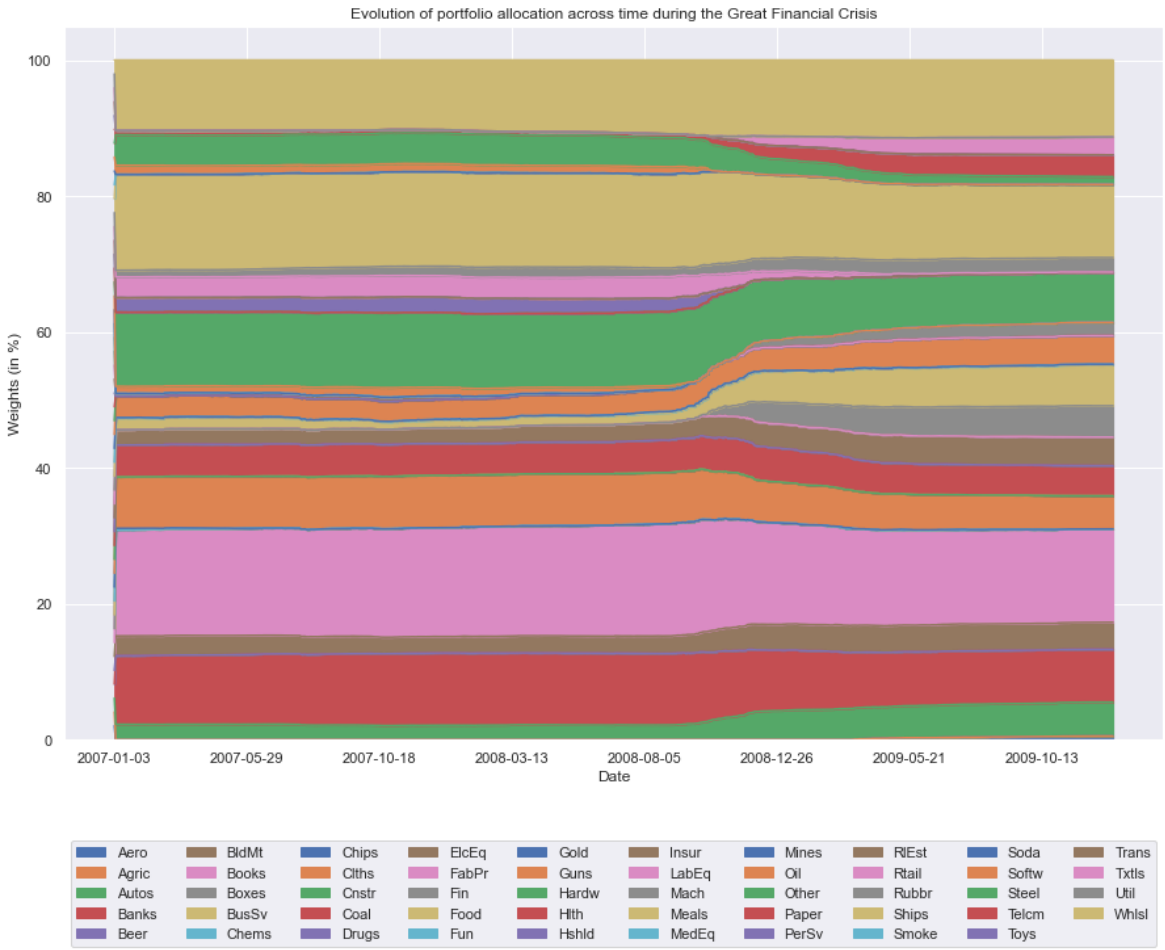

Figure 16: Evolution of portfolio allocation across time during the Global Financial Crisis

Upon further inspection, we identify the top 3 industries with the largest increase in portfolio allocation (winners) and top 3 industries with the largest decrease in portfolio allocation (losers). The change in allocation in these industries is represented by Figure 17. The vertical line represents 9/15/2008, the day Lehman Brothers filed for bankruptcy.

Figure 17 below validates our prior hypothesis to a reasonable extent. The DRL model leans into the Textiles, Wholesale, and Rubber industries as the crisis hits. Banks, Insurance firms, and Oil companies face the greatest decrease in allocation. Complete industry definitions can be found in Appendix 1.

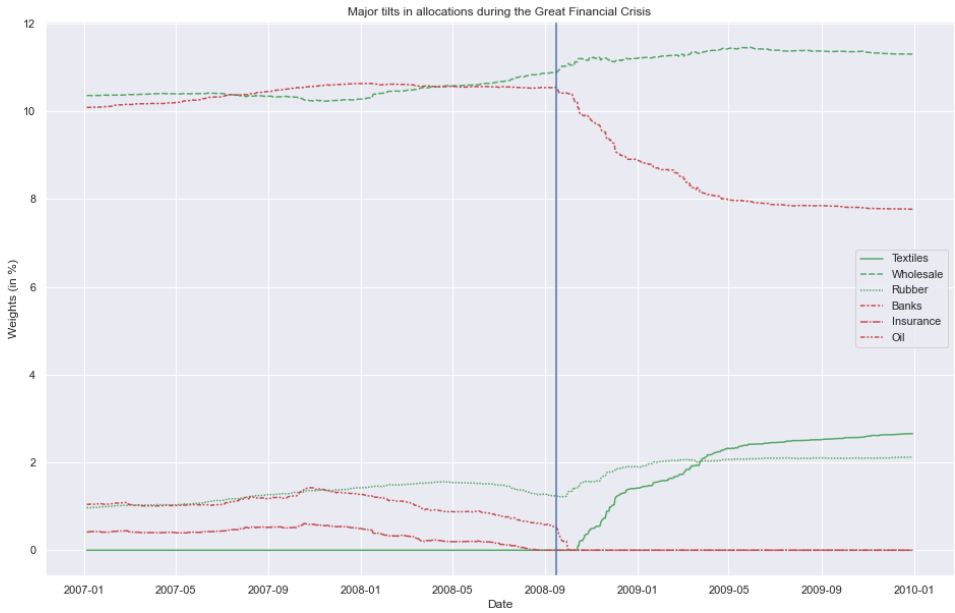

Figure 17: Tilt of portfolio allocation during the Global Financial Crisis

Further, we report summary statistics of the DRL strategy and compare it with the Minimum Variance Portfolio, and the S&P 500 in Table 7.

	DRL	Min Variance	S&P 500
Annual Return	0.066	0.019	-0.074
Sharpe Ratio	0.370	0.198	-0.106
Max Drawdown	-0.553	-0.520	-0.568

Table 7: Summary Statistics for DRL, Min Variance, and S&P 500 during the Global Financial Crisis

We also compare the log cumulative returns for the DRL strategy with the S&P 500 returns for our trading period in Figure 18.

Figure 18: Cumulative Log Returns of DRL compared with S&P500 and Minimum Variance during the GFC

Finally, we compare drawdowns for the DRL strategy with the S&P 500 performance in Figure 19.

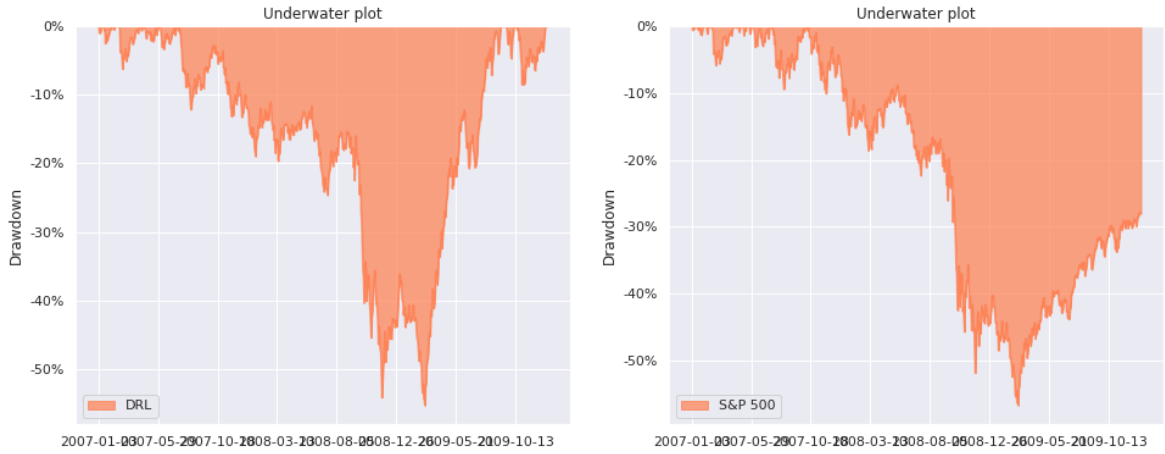

Figure 19: Drawdowns of DRL compared with S&P 500 during the GFC

To better understand our returns, we regress our them with the Fama French Carhart 4 factor Model and report our factor loadings in Table 8.

	Betas	P_vals
const	0.002	0.05
Mkt-RF	0.93	0.00
SMB	0.70	0.00
HML	0.11	0.00
Mom	-0.20	0.00

Table 8: Fama French 4 factor loadings during the Global Financial Crisis

To summarize, we see that the DRL strategy outperforms the Minimum Variance portfolio and the S&P 500 on a risk adjusted return basis. However, this higher return comes at a cost which can be seen in the slightly elevated maximum drawdown suffered by the DRL strategy as compared to the Minimum Variance. From the Underwater plot, we can also see that the DRL portfolio recovers swiftly as compared to the S&P 500. When regressed against the 4 factor Carhart Model, the portfolio has high (albeit less than 1) positive and significant betas for the Market and Size factors. This is consistent with our result in Section 8.3.

8.3.2 2020 Covid Crash

The analysis in this section closely mirrors the metrics and graphs used in Section 8.3.1. To focus on the covid crash of 2020, we train the model from 01/01/1970 to 12/31/2018. Using this trained model, we trade from 01/01/2019 to 12/31/2021. By trading during this window, we capture the portfolio allocations leading up to the crash and how they change after the crisis hits. We expect the model to tilt the portfolio allocation away from under performing industries. After running our analysis, we arrive at the time series of weights shown in Figure 20

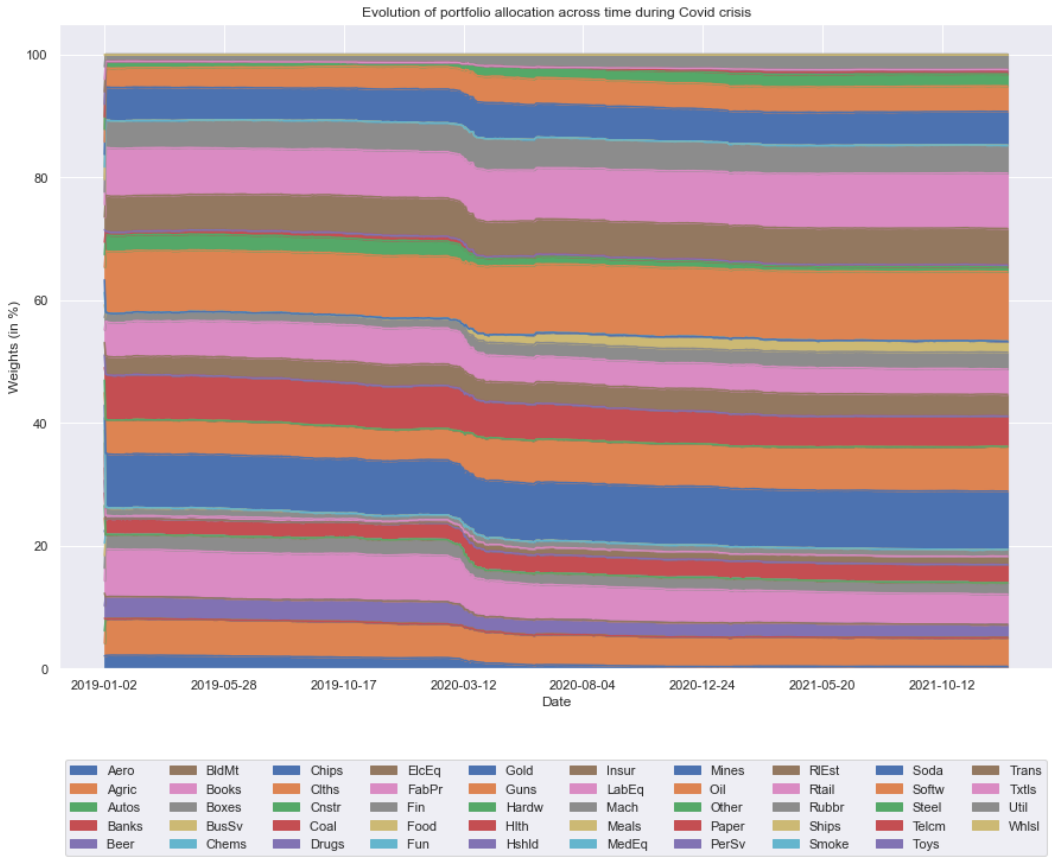

Figure 20: Evolution of Portfolio Weights across time during the Pandemic

As in section 8.4.1, we identify the top 3 industries with the largest increase in portfolio allocation (winners) and top 3 industries with the largest decrease in portfolio allocation (losers). The change in allocation in these industries is represented by Figure 21.

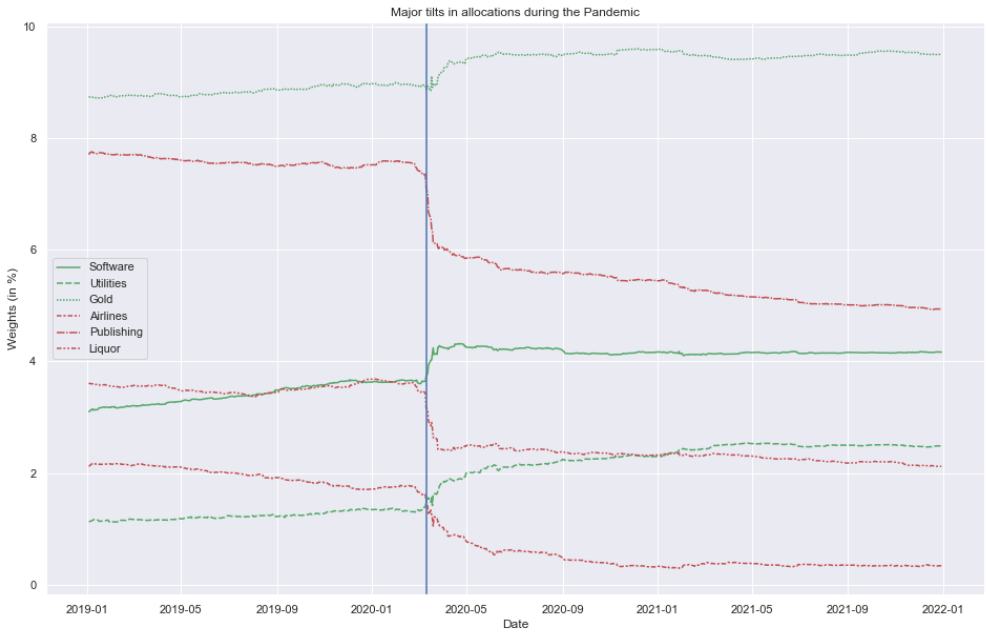

Figure 21: Tilt of portfolio allocation in the crisis

The above graph confirms our prior hypothesis that the DRL model tilts the portfolio weights away from the under-performing industries namely Airlines, Publishing, and Liquor. The model invests more into Software, Utilities, and Gold. We report the summary statistics of the DRL strategy and compare it with the minimum variance portfolio, and the S&P 500 in Table 9.

	DRL	Min Variance	S&P 500
Annual Return	0.320	0.254	0.239
Sharpe Ratio	1.153	1.152	1.069
Max Drawdown	-0.411	-0.350	-0.339

Table 9: Summary Statistics for DRL, Min Variance, and S&P 500 during the Pandemic

We also compare the log cumulative returns for the DRL strategy with S&P performance for this period in Figure 22.

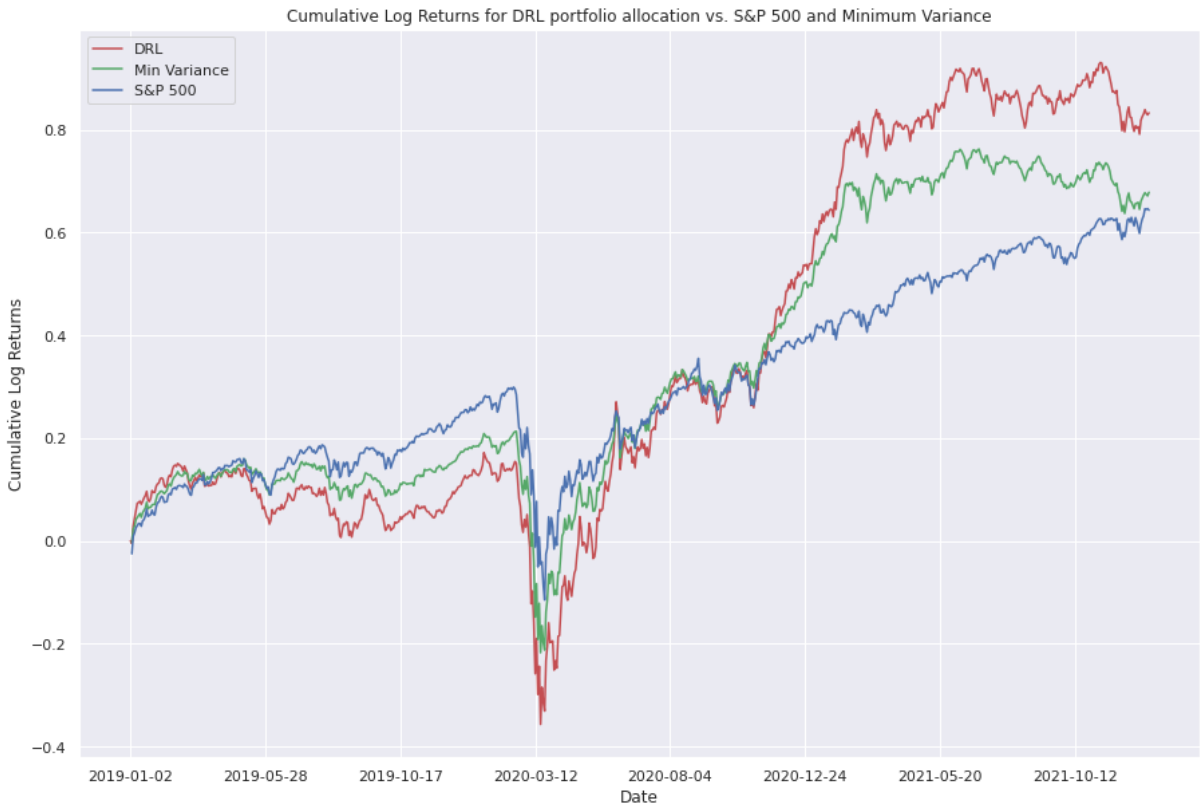

Figure 22: Cumulative Log Returns of DRL compared with S&P 500 and Minimum Variance

Finally, we plot the drawdowns for the DRL strategy and compare it with the S&P 500 in Figure 23

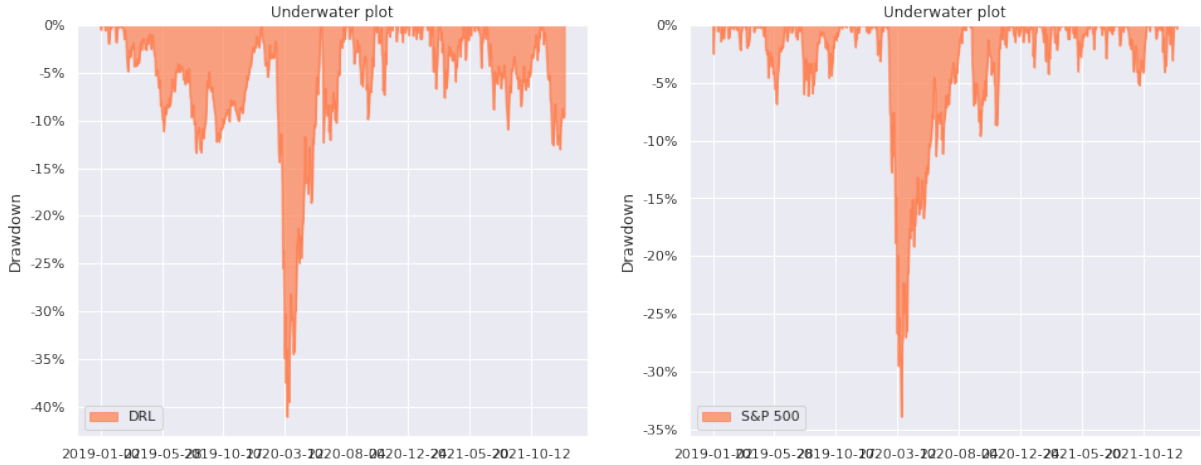

Figure 23: Drawdowns of DRL compared with S&P 500

In summary, we see that the DRL approach outperforms the S&P 500 and the Minimum Variance Portfolio on an absolute return basis. Looking at it from a risk-adjusted Sharpe lens,

DRL and the Minimum Variance portfolio comfortably beat the S&P 500. To understand these results better, we calculate the factor loadings on the Fama French 4 factor model for the DRL returns and report our results in Table 10. According to these loadings we see that our model is exploiting the size premium and over weighting small cap stocks. The S&P 500 by construction over weights large cap stocks and thus under performs. These results are consistent with our findings in Section 8.3.

	Betas	P_values
const	0.001	0.03
Mkt-RF	0.92	0.00
SMB	0.76	0.00
HML	0.31	0.00
Mom	-0.03	0.08

Table 10: Fama French 4 factor loadings during the Pandemic

9 Conclusion

9.1 Summary

Throughout this paper, we try to check the robustness of the DRL model and how it stands against the alternative portfolio allocation strategies. Instead of adding various features like the historical prices, covariance matrix, and other market features all at once in the model, we built the final model in three incremental approaches to quantify and attribute the enhancement of our model to these changes. This helped us measure the sensitivity of the model to different parameters. We also did a factor exposure analysis, but like in any other long-only portfolio, which is not market-cap weighted, the exposures were high to "Mkt-Rf" and "SMB". However, we do see some residual positive alpha which is significantly different from zero.

But like any other financial model, the true test of the model comes when it is put through a crisis period and we saw that DRL outperforms the alternative models significantly.

From our analysis, we conclude that if the DRL model is trained carefully and fed with relevant and clean data, it is a very strong contender for portfolio optimization, and it could potentially replace all of the traditional portfolio optimization methods.

9.2 Future Work

We have demonstrated that Deep Reinforcement Learning can solve the problem of portfolio allocation. To further our research, including risk in our reward function seems like a natural starting point. Currently, our reward function maximizes daily portfolio value. We can convert that into a function which maximizes return while penalizing big draw-downs.

Secondly, we could experiment with different agent architectures and see how the model behaves.

Given that we use Ken French's industry portfolios, we do not have any kind of information for the average daily volume traded. This information is crucial for calculating the transaction cost/market impact, hence we are taking a blanket cost of 10bps which is a fair estimate for our initial portfolio value. However, as part of future work the transaction cost could be modelled into this framework to get a more accurate estimate.

As we have shown in our analysis, adding information about market variables enhances the returns of our portfolio. We could extend the current model to incorporate information/signals from alternative datasets for our assets to further enhance the returns from portfolio optimization.

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Appendix 1

Fama French Industry definitions

Name in Paper	Long Name	Constituents
1 Agric	Agriculture	Agricultural production - crops
		Agricultural production - livestock
		Agricultural services
		Commercial fishing
		Prepared feeds for animals
2 Food	Food Products	Food and kindred products
		Meat products
		Dairy products
		Canned and preserved fruits and vegetables
		Flour and other grain mill products
		Bakery products
		Sugar and confectionery products
		Fats and oils
		Misc food preparations and kindred products
		Roasted coffee
		Misc food preparations
3 Soda	Candy and Soda	Candy and other confectionery
		Bottled-canned soft drinks
		Flavoring syrup
		Potato chips
		Manufactured ice
4 Beer	Beer and Liquor	Beverages
		Malt beverages
		Malt
		Wine
		Distilled and blended liquors
5 Smoke	Tobacco Products	Tobacco products
6 Toys	Recreation	Fishing, hunting and trapping
		Household audio visual equipment
		Phonograph records
		Boat building and repairing
		Musical instruments
		Toys
7 Fun	Entertainment	Services - motion picture production and distribution
		Services - motion picture theaters
		Services - video rental

Name in Paper	Long Name	Constituents
		Services - amusement and recreation
		Services - dance studios
		Services - bands, entertainers
		Services - bowling centers
		Services - professional sports
		Amusement and recreation services
		Services - Misc entertainment
8 Books	Printing and Publishing	Printing publishing and allied
		Newspapers: publishing-printing
		Periodicals: publishing-printing
		Books: publishing-printing
		Misc publishing
		Greeting card
		Bookbinding
		Service industries for the print trade
9 Hshld	Consumer Goods	Dog and cat food
		Curtains, home furnishings
		Household furniture
		Misc furniture and fixtures
		Soap and other detergents
		Perfumes, cosmetics and other toilet preparations
		Luggage
		Handbags and purses
		Personal leather goods, except handbags and purses
		Leather goods
		Pressed and blown glass
		Pottery and related products
		China and earthenware table articles
		Pottery products
		Glass products
		Household appliances
		Motorcycles, bicycles and parts
		Misc instruments, photo goods and watches
		Photographic equipment (Kodak etc, but also Xerox)
		Watches, clocks and parts
		Jewelry, precious metals
		Silverware
		Jewelers' findings and materials
		Costume jewelry and novelties
		Brooms and brushes

Name in Paper	Long Name	Constituents
		Burial caskets
10 Clths	Apparel	Apparel and other finished products
		Rubber and plastics footwear
		Leather tanning and finishing
		Boot and shoe cut stock and findings
		Footwear, except rubber
		Leather gloves and mittens
		Fasteners, buttons, needles, pins
11 Hlth	Healthcare	Services - health
12 MedEq	Medical Equipment	X-ray, electromedical app
		Surgical, medical, and dental instruments and supplies
		Ophthalmic goods
13 Drugs	Pharmaceutical Products	Drugs
		Biological products
		Medicinal chemicals
		Pharmaceutical preparations
		In vitro, in vivo diagnostic substances
		Biological products, except diagnostic substances
14 Chems	Chemicals	Chemicals and allied products
		Industrial inorganic chemicals
		Plastic material and synthetic resin/rubber
		Paints
		Industrial organic chemicals
		Agriculture chemicals
		Misc chemical products
15 Rubbr	Rubber and Plastic Products	Reclaimed rubber
		Rubber and plastic hose and belting
		Gaskets, hoses, etc
		Fabricated rubber products
		Misc rubber products
		Misc plastic products
		Misc rubber and plastic products
16 Txtls	Textiles	Textile mill products
		Floor covering mills
		Yarn and thread mills
		Misc textile goods
		Non-woven fabrics
		Cordage and twine
		Misc textile products

Name in Paper	Long Name	Constituents
		Textile bags, canvas products
		Misc textile products
17 BldMt	Construction Materials	Forestry
		Lumber and wood products
		Wood buildings and mobile homes
		Misc wood products
		Building paper and board mills
		Paving and roofing materials
		Stone, clay, glass, concrete, etc
		Flat glass
		Cement, hydraulic
		Structural clay products
		Vitreous china plumbing fixtures
		Porcelain electrical supplies
		Concrete, gypsum and plaster products
		Cut stone and stone products
		Abrasive and asbestos products
		Misc nonmetallic mineral products
		Cutlery, hand tools and general hardware
		Heating equipment and plumbing fixtures
		Fabricated structural metal products
		Metal doors, frames
		Architectural or ornamental metal work
		Prefabricated metal buildings and components
		Misc structural metal work
		Screw machine products
		Bolts, nuts, screws, rivets and washers
		Misc fabricated metal products
		Hard surface floor coverings
18 Cnstr	Construction	Build construction - general contractors
		General building contractors - residential
		Operative builders
		General building contractors - non-residential
		Heavy construction - not building contractors
		Construction - special contractors
19 Steel	Steel Works Etc	Primary metal industries
		Blast furnaces and steel works

Name in Paper	Long Name	Constituents
		Iron and steel foundries
		Primary smelting and refining of nonferrous metals
		Secondary smelting and refining of nonferrous metals
		Rolling, drawing and extruding of nonferrous metals
		Nonferrous foundries and casting
		Steel works etc
		Misc primary metal products
20 FabPr	Fabricated Products	Fabricated metal, except machinery and trans eq
		Fabricated plate work
		Sheet metal work
		Metal forgings and stampings
		Coating, engraving and allied services
21 Mach	Machinery	Engines and turbines
		Farm and garden machinery and equipment
		Construction, mining and material handling machinery and equipment
		Construction machinery and equipment
		Mining machinery and equipment, except oil field
		Oil and gas field machinery and equipment
		Elevators and moving stairways
		Conveyors and conveying equipment
		Cranes, hoists and monorail systems
		Machinery
		Metalworking machinery and equipment
		Special industry machinery
		General industrial machinery and equipment
		Refrigeration and service industry machinery
		Automatic vending machines
		Commercial laundry and dry-cleaning machines
		Air conditioning, warm air heating and refrigeration equipment
		Measuring and dispensing pumps
		Service industry machinery
		Misc industrial and commercial equipment and machinery

Name in Paper	Long Name	Constituents
22 ElcEq	Electrical Equipment	Electronic and other electrical equipment
		Electric transmission and distribution equipment
		Electrical industrial apparatus
		Electrical industrial apparatus
		Electric lighting and wiring equipment
		Residential electric lighting fixtures
		Commercial, industrial and institutional electric lighting fixtures
		Misc lighting equipment
		Communications equipment
		Misc electrical machinery and equipment
		Storage batteries
		Misc electrical machinery, equipment and supplies
23 Autos	Automobiles and Trucks	Tire cord and fabric
		Automotive trimmings, apparel findings and related products
		Tires and inner tubes
		Industrial trucks, tractors, trailers and stackers
		Vehicular lighting equipment
		Electrical equipment for internal combustion engines
		Transportation equipment
		Motor vehicles and motor vehicle equipment
		Motor vehicles and passenger car bodies
		Truck and bus bodies
		Motor vehicle parts and accessories
		Truck trailers
		Motor homes
		Travel trailers and campers
		Misc transportation equipment
		Misc transportation equipment
24 Aero	Aircraft	Aircraft and parts
		Aircraft
		Aircraft engines and engine parts
		Aircraft parts
		Misc aircraft parts and auxiliary equipment

Name in Paper	Long Name	Constituents
25 Ships	Shipbuilding, Railroad Equipment	Ship building and repairing
		Railroad Equipment
26 Guns	Defense	Guided missiles and space vehicles and parts
		Tanks and tank components
		Ordnance and accessories
27 Gold	Precious Metals	Gold and silver ores
28 Mines	Non-Metallic and Industrial Metal Mining	Metal mining
		Iron ores
		Copper ores
		Lead and zinc ores
		Bauxite and other aluminum ores
		Ferroalloy ores
		Mining
		Metal mining services
		Misc metal ores
		Anthracite mining
		Mining and quarrying nonmetallic minerals
29 Coal	Coal	Bituminous coal and lignite mining
30 Oil	Petroleum and Natural Gas	Oil and gas extraction
		Crude petroleum and natural gas
		Natural gas liquids
		Petroleum and natural gas
		Petroleum and natural gas
		Oil and gas field services
		Drilling oil and gas wells
		Oil and gas field exploration services
		Misc oil and gas field services
		Petroleum refining
		Misc products of petroleum and coal
31 Util	Utilities	Electric, gas and sanitary services
		Electric services
		Natural gas transmission
		Natural gas transmission and distribution
		Natural gas distribution
		Electric and other services combined
		Gas and other services combined
		Misc combination utilities

Name in Paper	Long Name	Constituents
		Water supply
32 Telcm	Communication	Communications
		Telephone communications
		Telegraph and other message communication
		Radio and TV broadcasters
		Cable and other pay TV services
		Communications
		Communication services (Comsat)
		Cable TV operators
		Telephone interconnect
		Misc communication services
33 PerSv	Personal Services	Rooming and boarding houses
		Camps and recreational vehicle parks
		Services - personal
		Services - laundry, cleaning and garment services
		Services - diaper service
		Services - coin-operated cleaners, dry cleaners
		Services - carpet and upholstery cleaning
		Services - Misc laundry and garment services
		Services - photographic studios, portrait
		Services - beauty shops
		Services - barber shops
		Services - shoe repair shops and shoeshine parlors
		Services - funeral service and crematories
		Services – Misc
		Services - tax return
		Services - Misc
		Services - photofinishing labs (School pictures)
		Services - auto repair, services and parking
		Services - automobile parking
		Services - automotive repair shops
		Services - automotive services, except repair (car washes)
		Services - Misc repair services
		Services - Electrical repair shops
		Services - Radio and TV repair shops

Name in Paper	Long Name	Constituents
		Services - Refrigeration and air conditioning service and repair shops Services - Electrical and electronic repair shops Services - Watch, clock and jewelry repair Services - Reupholster and furniture repair Services - Misc repair shops and related services Services - legal Services - educational Services - social services Services - museums, art galleries, botanical and zoological gardens Services - membership organizations Services - private households Services - truck and auto rental and leasing
34 BusSv	Business Services	Commercial printing Signs and advertising specialties Services - industrial launderers Services - business services Services - advertising Services - consumer credit reporting agencies, collection services Services - mailing, reproduction, commercial art and photography Services - services to dwellings and other buildings Services - building cleaning and maintenance Services - Misc equipment rental and leasing Services - medical equipment rental and leasing Services - heavy construction equipment rental and leasing Services - equipment rental and leasing Services - personnel supply services Services - computer programming and data processing Services - computer processing, data preparation and processing Services - information retrieval services Services - computer facilities management service

Name in Paper	Long Name	Constituents
		Services - computer rental and leasing
		Services - computer maintenance and repair
		Services - computer related services
		Services - Misc business services
		Services - security
		Services - news syndicates
		Services - photofinishing labs
		Services - telephone interconnect systems
		Services - Misc business services
		Services - RandD labs
		Services - management consulting and P.R.
		Services - detective and protective (ADT)
		Services - equipment rental and leasing
		Services - trading stamp services
		Services - commercial testing labs
		Services - business services
		Services - utility trailer and recreational vehicle rental
		Services - engineering, accounting, research, management
		Services - engineering, accounting, surveying
		Services - accounting, auditing, bookkeeping
		Services - research, development, testing labs
		Services - management, public relations, consulting
		Services - Misc
		Services - Misc engineering and architect
		Services - Misc
		Public warehousing and storage
35 Comps	Computers	Computer and office equipment
		Computers
		Computers - mini
		Computers - mainframe
		Computers - terminals
		Computers - disk and tape drives
		Computers - optical scanners
		Computers - graphics
		Computers - office automation systems

Name in Paper	Long Name	Constituents
36 Chips	Electronic Equipment	Computers - peripherals Computers - equipment Magnetic and optical recording media Computer integrated systems design Industrial controls Telephone and telegraph apparatus Communications equipment Radio and TV broadcasting and communications equipment Search, navigation, guidance systems Training equipment and simulators Alarm and signaling products Communication equipment Electronic components and accessories Search, detection, navigation, guidance Search, detection, navigation, guidance
37 LabEq	Measuring and Control Equipment	Engr lab and research equipment Measuring and controlling equipment Laboratory apparatus and furniture Automatic controls for regulating residential and commercial environments and applia Industrial measurement instruments and related products Totalizing fluid meters and counting devices Instruments for measuring and testing of electricity and electrical instruments Lab analytical instruments Optical instruments and lenses Misc measuring and controlling devices Optical instruments and lenses
38 Paper	Business Supplies	Office furniture and fixtures Paper and allied products Paper and allied products Manifold business forms Pens, pencils and other artists' supplies
39 Boxes	Shipping Containers	Wood containers Paperboard containers, boxes, drums, tubs Glass containers

Name in Paper	Long Name	Constituents
		Metal cans and shipping containers
40 Trans	Transportation	Railroads, line-haul operating
		Railway express service
		Local and suburban transit and interurban highway passenger transportation
		Local and suburban passenger transportation
		Taxicabs
		Intercity and rural bus transportation (Greyhound)
		Bus charter service
		School buses
		Motor vehicle terminals and service facilities
		Misc transit and passenger transportation
		Trucking and warehousing
		Trucking and courier services, except air
		Terminal and joint terminal maintenance
		Transportation
		Water transport
		Air transportation
		Pipelines, except natural gas
		Transportation services
		Freight forwarding
		Arrangement of passenger transportation
		Arrangement of transportation of freight and cargo
		Rental of railroad cars
		Misc services incidental to transportation
		Inspection and weighing services
		Packing and crating
		Misc fixed facilities for vehicles
		Motor vehicle inspection
		Misc transportation services
41 Whlsl	Wholesale	Wholesale - durable goods
		Wholesale - automotive vehicles and automotive parts and supplies
		Wholesale - furniture and home furnishings
		Wholesale - lumber and construction materials

Name in Paper	Long Name	Constituents
		Wholesale - professional and commercial equipment and supplies Wholesale - photographic equipment and supplies Wholesale - office equipment Wholesale - computers and peripheral equipment and software Wholesale - commercial equipment Wholesale - medical, dental and hospital equipment Wholesale - ophthalmic goods Wholesale - professional equipment and supplies Wholesale - metals and minerals, except petroleum Wholesale - electrical goods Wholesale - electrical apparatus and equipment Wholesale - electrical appliance, TV and radio sets Wholesale - electronic parts and equipment Wholesale - hardware, plumbing and heating equipment Wholesale - machinery, equipment and supplies Wholesale - machinery and equipment () Wholesale - construction and mining machinery andequipment Wholesale - farm and garden machinery and equipment Wholesale - industrial machinery and equipment Wholesale - industrial supplies Wholesale - service establishment machinery and equipment () Wholesale - transportation equipment, except motor vehicles Wholesale - Misc durable goods Wholesale - sporting goods and toys Wholesale - scrap and waste materials Wholesale - jewelry, watches, precious stones and metals Wholesale - durable goods Wholesale - nondurable goods Wholesale - paper and paper products

Name in Paper	Long Name	Constituents
		Wholesale - drugs and drug proprietaries Wholesale - apparel, piece goods and notions Wholesale - groceries and related products Wholesale - farm product raw materials Wholesale - chemicals and allied products Wholesale - petroleum and petroleum products Wholesale - beer, wine and distilled alcoholic beverages Wholesale - Misc nondurable goods
42 Rtail	Retail	Retail - retail-building materials, hardware, garden supply Retail - lumber and other building materials Retail Retail - paint, glass and wallpaper stores Retail - hardware stores Retail - nurseries, lawn and garden supply stores Retail - mobile home dealers Retail - general merchandise stores Retail - department stores Retail - general merchandise stores () Retail - variety stores Retail - catalog showroom Retail Retail - Misc general merchandise stores Retail - food stores Retail - grocery stores Retail - convenience stores Retail - meat and fish markets Retail - fruit and vegetable markets Retail - candy, nut and confectionary stores Retail - dairy products stores Retail - bakeries Retail - Misc food stores Retail - automotive dealers and gas stations Retail - automotive dealers

Name in Paper	Long Name	Constituents
		Retail - automotive and home supply stores Retail - gasoline service stations Retail - boat dealers Retail - recreation vehicle dealers Retail - motorcycle dealers Retail - automotive dealers Retail - apparel and accessory stores Retail - home furniture and equipment stores Retail - home furnishings stores Retail - household appliance stores Retail - radio, TV and consumer electronic stores Retail - computer and computer software stores Retail - record and tape stores Retail - musical instrument stores Retail Retail - Misc Retail - drug and proprietary stores Retail - liquor stores Retail - used merchandise stores Retail - Misc Retail - sporting goods stores and bike shops Retail - book stores Retail - stationery stores Retail - jewelry stores Retail - hobby, toy and game shops Retail - camera and photographic supply stores Retail - gift, novelty and souvenir shops Retail - luggage and leather goods stores Retail - sewing and needlework stores Retail Retail - non-store retailers (catalogs, etc) Retail Retail - fuel dealers and ice stores (Penn Central Co) Retail - Misc retail stores Retail - florists Retail - tobacco stores and stands Retail - newsdealers and news stands

Name in Paper	Long Name	Constituents
		Retail - optical goods stores
		Misc retail stores
43 Meals	Restaurants, Hotels, Motels	Retail - eating places
		Restaurants, hotels, motels
		Eating and drinking places
		Hotels and other lodging places
		Hotels and motels
		Membership hotels and lodging houses
		Services - linen supply
44 Banks	Banking	Depository institutions
		Federal reserve banks
		Commercial banks
		National commercial banks
		State commercial banks - Fed Res System
		State commercial banks - not Fed Res System
		National commercial banks - Fed Res System
		National commercial banks - not Fed Res System
		National commercial banks, not FDIC
		Misc commercial banks
		Savings institutions
		Banks ()
		Credit unions
		Foreign banks
		Functions related to depository banking
		Non-depository credit institutions
		Federal credit agencies
		FNMA
		SandLs
		Agricultural credit institutions
		Personal credit institutions (Beneficial)
		Business credit institutions
		Mortgage bankers and brokers
		Finance lessors
		Financial services
45 Insur	Insurance	Insurance
		Life insurance
		Accident and health insurance
		Fire, marine and casualty insurance
		Surety insurance

Name in Paper	Long Name	Constituents
		Title insurance
		Pension, health and welfare funds
		Misc insurance carriers
		Insurance agents, brokers and service
46 RlEst	Real Estate	Real estate
		Real estate operators and lessors
		Operators - non-resident buildings
		Operators - apartment buildings
		Operators - other than apartment
		Operators - residential mobile home
		Lessors of railroad and real property
		Real estate
		Real estate agents and managers
		Real estate dealers
		Title abstract offices
		Land subdividers and developers
		Real estate
		Combined real estate, insurance, etc
47 Fin	Trading	Security and commodity brokers, dealers, exchanges and services
		Holding and other investment offices
		Holding offices
		Management investment offices, open-end
		Management investment offices, closed-end
		Unit investment trusts
		Face-amount certificate offices
		Unit investment trusts, closed-end
		Trusts
		Investment offices
		Misc investing
		Oil royalty traders
		Commodity traders
		Patent owners and lessors
		Mineral royalty traders
		REIT
		Investors, NEC
48 Other	Almost Nothing	Sanitary services
		Steam and air conditioning supplies
		Irrigation systems
		Cogeneration - SM power producer